BIO303 FINAL TERM MCQs

.....Cycle allow seeds to use lipids source. **(glycoxylate)**

ΔS has a +ve sign when entropy..... **(increase)**

If ΔG is negative the reaction will occur..... spontaneously)

Equilibrium constants are..... **(multiplicative)**

Compounds that contain two phosphate or Phosphoryl groups attached at different positions In the molecule are named **bisphosphates**

FMN and FAD co enzymes catalyze oxidation Reduction reactions .....(**Flavoproteins**

)7) During preparatory phase..... Molecules of ATP Are invested before the cleavage of glucose into 2, **Three carbon compounds. (2)**

Aspergillus Niger and Lactobacillus are the Microbes used to commercially produce citric acid And lactic acid. The production take place in..... **(batch fermenter)**

In eukaryote TCA cycle takes place in .....**mitochondria**

The rate of any reaction is determined by the Concentration of the..... **(reactants)**

RNA is a polymer of..... **(ribonucleotides)**

Primary Structure of RNA defined as the Number and sequence of in the **RNA chain**.

Pepsin is a digestive enzymes in the stomach is Maximally active ph**.....(2)**

Ligand binds at a site on the protein called the... **(binding site)**

Mannose 6-phosphate is isomerized by Phosphomannose isomerase to yield fructose **6-Phosphate, an intermediate of glycolysis.**

ate equation V° is the.... Reaction velocity. **(initial)**

The velocity of reaction is the rate of .... Formation. **(product)**

Reversible inhibitors bind to enzyme through ..... Bonds. **(Non covalent)**

The ..... Inhibition in effects lower the Concentration of the active site. **(noncompetitive)**s

Neutral lipid are stored in ..... **(Adipocytes)**

The starting point for either the forward or the Reverse reaction is called the... **(ground state.)**

The force driving the system toward equilibrium Is defined as the..... **(standard free-energy Change)**

The substrate concentration at which V_0 is..... Is Km**____(½ Vmax)**

Since Vmax and Km are both constants, their Ratio is .... (**constant)**

What will be the pH of 0.1 M HCl?___**(1)**

At the beginning of the titration, the acetic acid Is..... ionized. **(Slightly)**

Citric Acid Cycle discovered by... **(Hans28)** Pyruvate, the product of glycolysis, is converted To... **(acetyl-CoA)**

29) Fatty acid entry into ..... Via the acyl-carnitine/ Carnitine transporter**. (Mitochondria)**

30) When a substrate attach to an enzyme is Causes... changes. **(conformational)**

31) ..... Reflects the affinity of enzyme for substrate. **(Km)**

32) When the enzyme is activated by DE phosphorylation **(by a specific phosphatase),**

Is citrate enters the citric acid cycle and ATP is Produced.

33) Fatty acid consist of an alkyl chain and....34) The solution is said to be Neutral pH when

There are exactly equal concentrations of positive And negative ions.

35) Chylomicrons transport dietary ..... to tissues. **(cholesterol)**

36) In citric acid cycle the NADH is released by the Formation of ..... **(oxaloacetate)**

37) Which coenzyme is not used in pyruvate Dehydrogenase complete. **(GTP)**

38) Energetic cost of activating fatty acid is Equivalent to .... Molecules of ATP. **(2)**

39) In general, irreversible inhibitors bind to Enzymes through **covalent bonds.**

40) Carboxylation of pyruvate by CO2 to form Oxaloacetate, catalyzed by **pyruvate carboxylase**

.41) ..... Serves as the universal energy currency in All living cell___**ATP**

42) Carbon in the fermentation medium .. can be Obtained from.**Upstream Processing**

43) Glycolysis and gluconeogenesis Are_____**Reciprocally regulated**

44) In nonoxidative phase, ribulose 5-phosphate is First epimerized to xylulose **5- Phosphate**

45) DNA forms RNA by transcription whereas the Process by which RNA form DNA is called **Reverse transcription**

46) Lyasis are the enzyme which do not catalyzed The reaction by **oxidation.**

47) A ligand bind at a site on the **protein**

48) The noncompetitive inhibition is effects the Concentration of the **active enzyme.**

49) Noncompetitive inhibitors do not interfere with **The binding of substrate to enzyme.**

50) Reversible inhibitors bind to enzyme through..... **Covalent Bonds.**

51) ..... Reflect the efficient of the enzyme for the Substrate. __**km**

52) Mass action ratio is depicted by..... **Q**

53) Neutral lipid are stored in ..... **adipocytes**

54) Fatty acid consist of an alkyl chain and ..... **Carboxylic acid**

55) Sine Vmax and Km are both constant **their ratio**

56) Enzyme..... System is described in terms of Free energy. **Biological System**

57) Ionization of water can be measured by it ..... **Electrical conductivity**

58)The rate or velocity of a reaction (v) is the Number of substrate molecules converted to Product per unit time; ___**Reaction Velocity.**

59) Oxidation of Odd-numbered fatty acids: **Fatty Acids with an odd number of carbons are found in The lipids of many plants and marine organisms**.

60) This cycle allows seeds to use lipids as a source Of energy to form the shoot during germination.__

Glyoxalate Cycle-2

61) If the non-protein moiety is a metal ion such as62) High effective catalyst commonly reaction rate

By a factor of .....10^{5} to 10^{17}

63) Reaction velocity denoted by..... **V**

64) The rate of any reaction is determined by the Concentration of the..... **reactants**

65) Pepsin a digestive enzymes in the stomach is Maximally active at pH..... **2**

66) Initial velocity is denoted by.... **V·**

67) Unequal electron sharing.....**two electric dipoles**

68) H30+ migrate towards the..... **Cathode**

69) . .....Cycle allow seeds to use lipids source. **(Glycoxylate cycle -2)**

70) ΔS has a +ve sign when entropy..... **(increase)**

71) If ΔG is negative the reaction will occur.....

72) Equilibrium constants are..... **(multiplicative)**

73) FMN and FAD co enzymes catalyze oxidation Reduction reactions**.....(Flavoproteins)**

74) Gluconeogenesis occurs in the **a.Plant b.Animals c.Microorganisms. d. All options**

75) Utilization of glucose the hydrolysis of ATP Produced the **ADP + pi**

76) Glycolysis is the____**Exergonic**

77) Beta oxidation__The repetitive four-step Process, by which fatty acids are converted into

Acetyl-CoA called Beta oxidation.

78) In eukaryote, TCA cycle takes place in **.....mitochondria**

79) The last step in glycolysis is the transfer of the Phosphoryl group from phosphoenolpyruvate to

ADP, catalyzed by pyruvate kinase, which Requires K+ (potassium ions) and either Mg+2 or Mn+2 .

80) The free energy of the system is plotted against **The progress of the reaction S → P.**

81) A buffer system consists of a weak acid **(the Proton donor)** and its conjugate base (**the proton**

82) If Keq of a reaction is greater than **1.0**, **ΔG˚** is **Negative**, forward direction is favored

83) If Keq is less than **1.0,** ΔG˚ is **positive**, **reverse** Direction is **favored**

84) Secondary structure involves coil formation of **The polyribonucleotide chain.**

85) 3-D arm: **It contains the base dihydrouridine.**

86) Km reflects the affinity of the enzyme for the **Substrate.**

87) Liver peroxisomes do not contain the enzymes Of the citric acid cycle and cannot catalyze the

Oxidation of acetyl-CoA to CO2.

88) Nucleophiles: **functional groups rich in and Capable of donating electrons**

89) ΔS has a **+ve** sign when entropy increase and **_--ve** when entropy decreases

90) In the intestinal mucosa they are reconverted to **Triacylglycerol's and packed with dietary**

Cholesterol and specific proteins into lipoprotein Aggregates called **chylomicrons**.

91) The concentration terms in this equation express The effects commonly called mass action, and the

Term [C]c[D]d/[A]a[B]b is called the mass-action Ratio, Q.

92) Too much alcohol drinking reduce niacin **Absorption from the intestine**

93) The binding of a protein and ligand is often Coupled to a conformational change in the

Protein that makes the binding site morecomplementary to the **ligand, permitting tighter**

Binding called induced fit.

94) The ionization of water can be measured by its **Electrical conductivity**

95) The first step in glycolysis is phosphorylation Of glucose at the hydroxyl group on C-6 by a

Family of enzymes called hexokinases to form

Glucose-6- phosphate (G6P). · step 2: The D-

Glucose 6-phosphate thus formed is converted to D-f

96) Chylomicrons **(transport dietary cholesterol to Tissues)**

97) The rate of an enzyme catalyzed reaction Increases with substrate concentration until a

98) Some rearrangement reactions, such as that Catalyzed by DNA photolyase__- **Free-Radical Reactions**

99) There is a linear relationship between reaction Rate and enzyme concentration **(at constant**

Substrate concentration).

100) Acetyl-CoA derived from fatty acids thus Enters a final common pathway of oxidation with

The acetyl-CoA derived from glucose via Glycolysis and pyruvate oxidation.

101) Enzyme is not rigid, changes shape with **Substrate**.

102) The substrate concentration at which V_0 is Half maximal is Km, the **Michaelis constant**.

103) This change in conformation when the Substrate binds is induced by multiple weak Interactions with the **substrate**

104) One simplifying approach in kinetics Experiments is to measure the initial rate (or Initial velocity), designated V0, when **[S] is much Greater than the concentration of enzyme, [E].**

105) In general, irreversible inhibitors bind to Enzymes through **covalent bonds.**

106) Reversible inhibitors typically bind to Enzymes through **non-covalent bond**

107) The PDH complex is composed of multiple Copies of three enzymes: · **Pyruvate**

Dehydrogenase, · E1, E2 and E3

108) When there are exactly equal concentrations Of H+ and OH- , as in pure water, the solution is

Said to be at neutral pH.

109) The noncompetitive inhibitor, in effect, Lowers the concentration of the active enzyme and

Therefore **decreases the Vmax of the enzyme**

110) The tendency of any acid (HA) to lose a Proton and form its conjugate base (A-) is defined

By the acid dissociation constant (Ka) for the Reversible reaction.

111) Buffer Solutions: Buffers are aqueous Systems that tend to resist changes in pH when

Small amounts of acid (H+) or base (OH-) are

112) Lineweaver-Burk plot has the great Advantage of allowing a more accurate **Determination of Vmax,**

113) “Warburg effect” The German biochemist Otto Warburg first observed in 1928 that Tumors_____**Cancer**

114) Fatty acids catabolism causes the oxidation Beta carbon to form___**Glycerol**.

115) ___Inhibitors binds at site disconnect from The substrate binging site __**Non-Competitive**

116) At the beginning of the titration acetic acid Is ___ionized___**Slightly**

117) In organic chemistry, the acryloyl is form of __**Enone**

118) The binding of enzyme to the sIte of __**Enzyme**

119) Fatty acids consists of an alkyl chain and__**carboxylic Acid**

120) Glyoxalate cycle is a variation of _**TCA Cycle.**

121) The value of G remains positive when System __Energy __**Gain**.

122) The payoff phase of the Glycolysis includes The energy containing step which involves ___**Phosphorylation**.

123) All tRNA molecules contain __main arms or Loops.__**4**

124) Delta S has a positive Sign when Entropy____**Increases**

125) Asspergillus niger and Lactobacillus are the Microbes used to commercially produce Citric acidand lactic acid. The production take place in __**Batch Fermenter.**

126) Which Coenzyme is not used in pyruvate Dehydrogenase complete___**GTP**

127) The Force driving the system toward the Equilibrium is defined as the __**Standard Free**

Energy change.

128) Carbon in the fermentation obtained from___**Beet Molasses**

129) In the cytosol, oxaloacetate is converted to Phosphoenolpyruvate by **PEP carboxykinase.**

130) The level of fructose 2,6-bisphosphate is high in **The fed state and low in starvation.**

131) the single strand of RNA is capable of **self-Folding like a hairpin**

132) The core of the Escherichia coli enzyme contains **24 copies of E2**

133) The German biochemist Otto Warburg first Observed in 1928 that tumors of nearly all types carry Out glycolysis____ "**Warburg effect"**

134) like the saturated fatty acids, enters the Mitochondrial matrix via the **carnitine shuttle**

135) RNA typically exists as a **single strand Globular Molecule**

136) Both glycolysis and gluconeogenesis are **Essentially irreversible processes in cells**

137) Therefore, at the end of two Citric acid cycles, **The products are: two GTP, six NADH, two FADH2,And four CO2.**

138) Oxidation of fatty acids to **acetyl-CoA**. This139) Acetyl-Coa formed from dietary fiber and Bind Coa Cycle__**Citric cycle**

140) acetyl-CoA formed in the liver during oxidation of fatty acids can either enter the citric acid cycle or undergo conversion to the **"ketone bodies,"**

141) energy in biological systems is described in terms of free energy, **G′º**

1. In animals TCA cycle take place in **Mitochondria**

2. Rate equation Vo is the ..... **Reaction velocity Initial**

3. Velocity of reaction is rate of ..... (is ki statement kuch differ thi) a. **Product**

4. What will be the PH of 0.1M HCL? **1**

5. At the beginning of titration ACETIC ACID is ..... ionized a. **Slightly**

6. Neutral lipids are stored in ..... **Adipocytes**

7. The starting point of either forward or reverse reaction called **Ground state**

8. Vmax and Km are both constant at **zero**

9. Reflects the affinity of enzyme for substrate is a. **Vmax**

10. line weaver Burke plot is determined by **Km**

11. In citric Acid cycle how many NADH are produced 3

12. In fatty acid formation how many ATP molecules are formed **4**

13. Exergonic process .....release.....**energy**

14. Formation of DNA from RNA by c**. Reverse transcription**

15. Mg+ cofactor mainly used as ..... process

a. **All biological process**

16. Equilibrium constant described by **Both k' or k' eq**

18. Commonly used name for most enzymes followed by suffix b. -**ase**

19. The Value of G remains +ve when it gain..... **energy**

20. Glyoxalate cycle is variation of **Tca**

21: The carnitine shuttles a mechanism by which long chain fatty acids are transported in to **Mitochondrial matrix**

22. Beta oxidation leads to **Fatty acid degranulation**

23. Reversible inhibitor binds to enzyme ..... bond **Non-covalent**

24. Chyrmnins transport dietary ..... Tissues. **Cholesterol**

25. In citric Acid Cycle NADH released by .....**Oxaloacetate**

142) .....are single strands globular molecules.___ **tRNA**

143) Animals convert the chemical energy of fuels into motion and heat and In a few organisms such as fireflies **chemical energy is converted into light**

144) A ligand bind at a site on the protein called the___**binding site**.

145) Gluconeogenesis is a process_____**Synthesis glucose**

146) Cori cycle take place in___**liver**

147) Conversion of glucose 6_phosphate of fructose 6_ phosphate requires enzymes_____**phosphohexose isomerase**

148) Standard reduction is denoted by____**E°**149) Pyruvate is oxidized, with loss of its carboxyl group as CO_2 , to yield the **acetyl group of acetyl-coenzyme A;**

150) Chemicals reaction in which the reactions and products same phase____**homologous chemical reactions**

151) Covalent bond are also called____**molecular bond**

152) It is in the form of a loop and carries specific sequences of three bases which **constitute the anticodon.**

153) Secondary structure involves coil formation of the **polyribonucleotide chain**.

154) It is defined as the number and sequence of ribonucleotides in the RNA chain._____ **Primary Structure of RNA**

155) Separation of the two strands of the double helix when hydrogen bonds between the paired bases are disrupted_____ **DNA Denaturation**

156) In non-oxidative phase, ribulose 5-phosphate is first epimerized to **xylulose 5phosphate**

157) In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by **PEPcarboxykinase**

.158) The important precursors of glucose in animals are three-carbon compounds such as **lactate, pyruvate, and glycerol**, as well as **certain amino acids**

159) We take the example of Aspergillus niger and Lactobacillus are the microbes used to commercially produce citric acid and lactic acid, respectively**. (Both Fermentation)**

160) Lactic acid fermentation is done by some fungi, some bacteria like in yogurt, and **sometimes by our muscles**.

161) Compounds with two phosphates linked together as a pyro phosphoryl group are named **diphosphates;**

162) Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named bisphosphates **(or bisphospho compounds);**

163) The energy gain comes in the payoff phase of **glycolysis**

164) Note that during preparatory phase, 2 molecules of ATP are invested before the cleavage of glucose into **two three-carbon pieces.**

165) NAD and NADP are soluble electron carriers that act with **dehydrogenase enzymes**

166) Waste products from other industrial processes, such as molasses, cheese whey and corn steep liquor, after modifying with **theincorporation** of additional nutrients, are used as the **substrate for many industrial fermentation**

167) · A high level of AMP indicates that the energy charge is **low**

168) High levels of ATP and citrate indicate that the energy charge is **high**

169) In 1910 **Chaim Weizmann (later to become the first president of Israel**) discovered that the bacterium Clostridium acetobutyricum ferments starch to **butanol and acetone.**

170) Biological Reactions required energy ____**endergonic.**

171) Actual free-Energy changes depend on **Reactant and Product Concentrations**

172) The binding of a protein and ligand is often coupled to a conformational change in the **protein**

173) All tRNA molecules contain **4** main arms or loops.174) Most biochemical reactions occur near pH **7 (buffered)**

175) beer and wine, alcohol are produced by **fermentation**

176) In some plant tissues and in certain invertebrates, protists, and microorganisms such as yeast, pyruvate is converted under hypoxic or

anaerobic conditions to ethanol and CO2, a process called ethanol **(alcohol)** fermentation

177) Under Aerobic conditions, the pyruvate formed in the final step of glycolysis is oxidized to acetate (acetyl- CoA), which enters the citric acid cycle and is **oxidized to CO2 and H2O.**

178) Dehydrogenase meaning ____**Removal of Hydrogen**

179) Galactokinase deficiency galactosemia high conc. are found in **blood and urine.**

180) Messenger RNA (mRNA) Size **500-6000 nucleotides**

181) The rate of re-association depends upon the

concentration of the complementary strands.182) Pyruvate is the end product of the second phase of

glycolysis

183) Tri carbocyclic acid cycle (TCA) or **Kreb's cycle**

184) Kreb's cycle same as **Acetyl Co-A**

185) The enzyme phosphohexose isomerase(phosphoglucose isomerase) catalyzes the reversible isomerization of glucose 6- phosphate, **an aldose, to fructose 6- phosphate, a ketose.**

186) Unshared electrons are represented by **Dot**.

187) Similar to DNA, RNA polymer also has **polarity**

188) Too much alcohol drinking **reduce niacin absorption from the intestine**

189) Oxalo acetic acid (**OAA**) produced in mitochondria is impermeable through **mitochondrial membrane**

190) Lipid-soluble quinones: ubiquinone and plastoquinone act as electron carriers and **proton donors in**

191) NAD is a lipid soluble **electro carriers.**

192) In the surface techniques, the microorganisms are cultivated on the surface of a **liquid or solid substrate.**

193) In the submersion processes, the microorganisms grow in a liquid medium provided with the **substrate**

194) Lyases are included in a separate class because they catalyze these reactions by means other than **hydrolysis or oxidation.**

195) Nucleophiles: **functional groups rich in and capable of donating electrons**

196) The electron-donating molecule in an oxidation-reduction reaction is called the **reducing agent or reductant**

197) The electron-accepting molecule is the **oxidizing agent or oxidant**.

198) Photosynthetic organisms make glucose by first reducing atmospheric **CO2 to trioses**, then **converting the**

199) NAD, NADP are **water soluble coenzymes**.

200) The payoff phase of glycolysis includes the energy- c**onserving phosphorylation steps**

201) RNA typically exists as a single strand Molecules whereas DNA exists as a double-**stranded helical molecule.**

202) The process of synthesis of glucose from non- carbohydrate sources is known as Gluconeogenesis**(abbreviated GNG)**

204) This reannealing process is also referred to as**205**) Since the RNA molecule is a single strand complementary to only one of **the two strands of a gene,**

206) miRNA is a type of **ncRNA**

207) When the amount of ATP is high in cell, glucose is converted into **Glycogen**.

208) Reactions in which system gains free energy is called **Endergonic**.

209) Phosphoenolpyruvate is an **Enol Form**.

210) A phosphoryl group (-PO3 2-) from ATP is transferred to an alcohol (**forming a phosphate ester)**